Ashmit Mukherjee

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Education

New York University Abu Dhabi *Expected May 2027*
B.S. in Computer Science GPA: 3.5
Study Away: NYU Paris; NYU College of Arts & Science (New York)

Relevant Coursework: Natural Language Processing; Principles of Data Science; Statistics for Social & Behavioral Sciences; Data Structures & Algorithms; Computer Systems Architecture; Operating Systems

Research Interests

I work at the intersection of natural language processing, machine learning, and computational social science. My current work spans multilingual and code-mixed NLP, protein representation learning, and multi-agent simulation of Human–AI interaction. More broadly, I am interested in how models learn structure from social and biological data — including questions of evaluation, interpretability, and what it means for a model to generalize meaningfully across languages, domains, and tasks.

Research Experience

Research Assistant — Human-AI Collaboration, NYU Abu Dhabi *Feb 2026–Present*

- Implementing multi-agent Human–AI social dilemma simulations in Python (NumPy, NetworkX) using game-theoretic and sequential decision-making frameworks.
- Modeling norm emergence and coordination dynamics under heterogeneous incentives; analyzing outcomes through simulation-based experiments and statistical evaluation.
- Developing AI-mediated intervention mechanisms and evaluating cooperative equilibria using PyTorch-based agents, reinforcement learning setups, and modular simulation pipelines.

Research Assistant, eBRAIN Lab — NYU Abu Dhabi *Jan 2026–Present*

- Researching scientific large language models and protein representation learning for membrane protein and signal peptide prediction tasks.
- Benchmarking pretrained protein language models (ESM, ProtBERT) across supervised biological classification settings using standardized datasets and evaluation metrics.
- Exploring parameter-efficient fine-tuning (LoRA) to inject physicochemical priors such as hydrophobicity while reducing training cost.
- Collaborating with doctoral researchers on experimental design, ablation studies, and reproducible evaluation pipelines.

Research Projects

Hinglish Named Entity Recognition Benchmark *Fall 2024*

- Fine-tuned multilingual transformer models (mBERT, XLM-RoBERTa) on the COMI-LINGUA dataset for Hindi–English code-mixed NER.
- Achieved 78% entity-level F1; conducted controlled benchmarking against zero-shot GPT-4o and LLaMA 3.1 baselines.
- Analyzed trade-offs between model scale, task-specific fine-tuning, generalization, and computational cost.

Data Science Salary Prediction Platform (Methodological Study) *Fall 2025*

- Conducted large-scale regression benchmarking using AutoML (PyCaret) across 15+ models on a dataset of 3,000+ job postings.
- Analyzed feature contributions using SHAP to study labor market dynamics across geography, firm size, and skill requirements.
- Tracked experiments with MLflow; reported MAE, RMSE, and R^2 across conditions.

Crime Data Analysis and Prediction *Spring 2025*

- Studied relationships between socioeconomic indicators and violent crime rates using regression modeling on data from 1,994 communities.

- Evaluated model performance via cross-validation ($R^2 = 0.85$) and exploratory statistical analysis; built interactive visualizations for interpretability.

CAMP — Campus Asset Management Platform *Fall 2024*

- Designed and implemented a multi-user system for resource allocation and scheduling under shared constraints.
- Developed RESTful APIs and automated CI/CD pipelines using Docker and GitHub Actions.

Technical Skills

Languages: Python, C++, C#, JavaScript
ML & NLP: PyTorch, Transformers (Hugging Face), Scikit-Learn, XGBoost, PyCaret, SHAP
Multi-Agent & RL: NumPy, NetworkX, reinforcement learning pipelines
Data Analysis: Pandas, SciPy, Matplotlib, Seaborn, Plotly
Experiment Tracking: MLflow, Git, Docker, GitHub Actions
Other: Streamlit, MongoDB, LaTeX

Additional Experience

Machine Learning Engineer Intern, Zeek *Jun–Aug 2025*

- Developed and evaluated segmentation models under noisy, real-world business constraints.
- Applied model interpretability techniques to study robustness and bias sensitivity.